



INFORMATION SHEET 9

WATER MANAGEMENT

Soil and water management are closely interrelated. Although there are great differences in the water requirements of different plants, no plant growth is possible without a certain amount of water. Out of necessity, however, a number of plants have developed great drought tolerance, and some even drought resistance.

Plants can contain up to 90 percent water. The water is absorbed mainly through the root system of the plant. With the water, plant nutrients are absorbed. Healthy roots need air (aeration) for development. Excess water in the soil prevents air from penetrating and damages a plant's roots. Water management is therefore extremely important in regions with good water resources as well as in those where water is scarce.

The water-holding capacity of soil varies according to soil type. Soil with a high content of organic matter has better aeration, better structure and better water-holding capacity.

Heavy, sticky soils are too dense to allow air in and water out, so roots cannot breathe and plants can have growth problems. When this kind of soil dries out, it sets like cement, and water takes a long time to soak into it. On the other hand, sandy, coarse-grained soils are too loose to hold water before it drains away. In this kind of soil, without a regular external water supply, a plant's roots cannot find enough water for growth. Regular application of organic matter will improve the ability of both these kinds of soil to hold and release enough water and air.

WATER MANAGEMENT IN HIGH RAINFALL AREAS

In high rainfall areas or wetlands, water management involves limiting the damage from temporary excess water. Drainage is therefore the most important factor. On sloping land, cut-off drains made along contour lines lead the water away. The gradient should not exceed 0.4 percent (i.e. a difference in elevation of 10 cm in 25 m) to slow the flow of water and avoid erosion. Water can be collected for later use in a hand-dug pit at the end of a cut-off drain.

A good way to manage excess water is to put plants that can tolerate more soil moisture near the water source (wetlands, the banks of a stream or a flooded area). Less water-tolerant plants can be planted on mounds or raised beds. If properly managed, some wetlands (which are known as *dambos* in Zambia, *mapani* in Zimbabwe, *mbugas* in the United Republic of Tanzania and *marais* in Rwanda) can be used to grow crops year round. Excess drainage should be avoided, especially on organic soils.

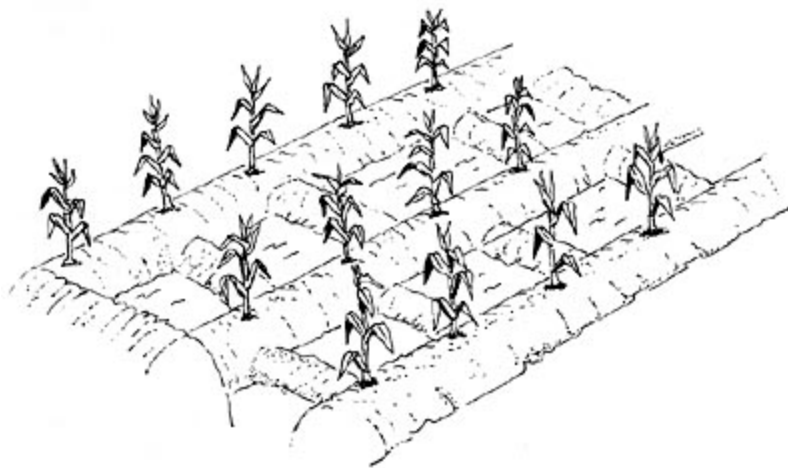
Low-lying wetlands are subject to seasonal or permanent flooding. This may be the result of surface runoff or groundwater seepage from a catchment area over an impermeable soil layer towards low-lying areas. In such areas, organic matter decomposes slowly, forming the black, mulch- and humus-rich topsoil that is associated with high soil fertility.

Wetlands are important for smallholder food production for a large part of the year, especially during the dry season. In East Africa, southern Africa and some northern parts of West Africa, home gardens are mainly found in such wetlands. Since the wetlands can remain wet for a considerable part of the year, most crops are grown on raised beds or ridges (also referred to as mounds) to reduce waterlogging and facilitate drainage. Ridges will keep excess water away from the plants and lead it directly to the plant's roots, as illustrated in Figure 1. When the water table falls, water is channelled into pits and scooped with tins on to the ridges. If the water table sinks farther, plants are grown in flat beds and, eventually, at the peak of the dry season, sunken beds. Also plants can be established on selected parts (top, side or bottom) of the ridge according to their water requirements and waterlogging tolerance.

Canals and trenches drain water from rain-flooded areas. Plants that require frequent watering, such as rice, sugar cane, taro and water spinach, can grow in them. Other plants, such as cassava, yam and leafy vegetables, can be grown on raised beds. Wetlands are delicate environments and should be used with great care. Both soil erosion and excessive drainage damage wetlands.

FIGURE 1

Ridges help keep water away from plants and lead it directly to the roots



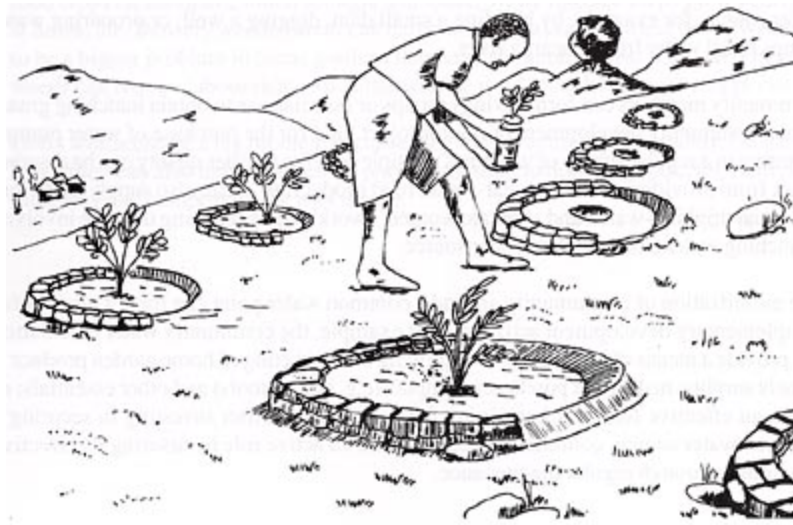
WATER MANAGEMENT IN A DRY CLIMATE OR DURING THE DRY SEASON

Maintaining soil moisture and collecting water from different sources is more difficult and labour intensive than keeping excess water off crops. Water management in dry areas requires efficient use of available water resources through the prevention of runoff and the reduction of evapo-transpiration, i.e. the prevention of loss of water through evaporation from the soil, and from the leaves of plants. Under dry conditions, plants should be placed to make optimum use of available moisture. It is, of course, most important to choose plants with drought tolerance and low water requirements. Deep-rooted plants, such as African eggplant, can usually survive periods of limited water availability better than shallow-rooted species. Once established, permanent crops, such as many trees and shrubs, will withstand reduced water conditions as well.

It is also important to make the best use of rainfall and make moisture available to plants for as long as possible. During the rainy season, basin-like or sunken beds are often used to keep a maximum amount of water accessible to plants and to prevent runoff of surface water. Microbasins have also been in use for a long time and are well suited to sloping land as well. They are built by placing a ring or semicircle of stones, soil or other material around individual plants, in particular tree crops. It is important not to heap the soil too high on the plant's stem, as this might damage the bark. Instead, water should be able to collect in a ring above the root line, some distance from the stem, as illustrated in Figure 2.

FIGURE 2

Microbasin around a tree



If plants suffer from direct sunlight and heat, water is lost through transpiration. This can be prevented or reduced by creating natural windbreaks and providing shade. Mulching will also reduce evaporation of moisture from the soil.

Weeds should be removed because they compete with cultivated plants for moisture. To reduce crop failure, short-term crops can be grown near the water source. High organic matter content in soil helps retain the soil's moisture. Compost and green manure help increase the organic matter content of soil. Mulching helps hold water in the soil by keeping the soil surface from drying out and becoming too hot. Light-coloured mulches, such as straw, are particularly good, since they reflect sunlight and heat away from the soil. See Home Garden Technology Leaflet 6, "Special techniques for improving soil and water management", for more details.

COMMUNITY MOBILIZATION AROUND A WATER SOURCE

In the humid and subhumid parts of Africa, crops grow under rainfed agriculture for about seven months of the year, and the demand for irrigation water is relatively low. In addition, the water table is fairly high during the rainy season; therefore, the cost of securing a reliable water source for home gardening purposes is also relatively low. For households in the semi-arid regions, however, the situation is quite different. Individual households in these areas often do not have access to a reliable year-round water source.

In such areas, the need for a reliable water source can become an effective driving force for mobilizing a community. This can be achieved in different ways. For example, households, in consultation with the relevant technical extension sectors of governmental or non-governmental agencies, can form groups and decide on the most appropriate water source to develop. Interested households can then pool their resources to establish a common water source, for example, by building a small dam, digging a well, or procuring water pumps to lift water from a nearby river.

Community members can form savings groups or associations to obtain matching grants from a community development or microproject fund for the purchase of water pumps. Investing in a regular supply of water has multiple benefits if water quality can be assured. Apart from providing water for year-round food production, it can also supply sufficient and clean drinking-water, and alleviate women's workload by reducing the time involved in fetching water from a distant water source.

The mobilization of a community around a common water point can form a nucleus for complementary development activities. For example, the community water association can provide a means of organizing or improving the marketing of home garden produce if there is surplus; make joint purchases of inputs (e.g. seeds, tools) and other essentials; or create an effective force for bargaining for better prices. After investing in securing a common water source, communities need to take an active role in ensuring its effective functioning through regular maintenance.

